National College of Ireland

Cloud Platform Programming

Application Integration/Messaging Services

Outline

- Introduction to Message Queues
- Amazon SQS
- Amazon SNS

Introduction to Message Queues

- A message queue – asynchronous service-to-service communication used in distributed computing, (micro)services, and serverless architectures
- Use cases:
  - decouple heavyweight processing
  - buffer or batch work
  - smooth spiky workloads

source: https://aws.amazon.com/message-queue/

Introduction to Message Queues

- Message queues allow different parts of a system to communicate and process operations asynchronously
- Message queues improve the performance, reliability and scalability of service oriented applications/ distributed applications

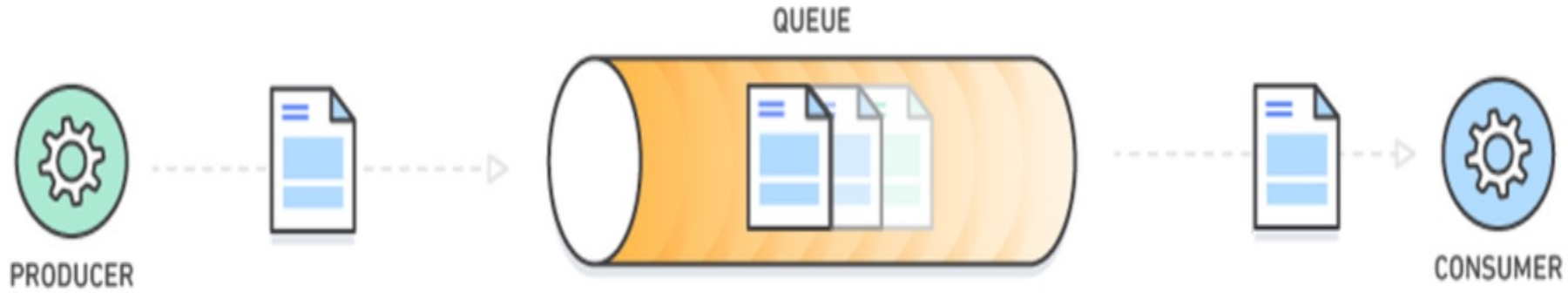
source: https://aws.amazon.com/message-queue/

Introduction to Message Queues

- A message queue provides
  - A buffer which temporarily stores messages
  - endpoints that allow software components to connect to the queue in order to send and receive messages
- The messages are usually small
  - e.g. requests, replies, error messages, or data/information

source: https://aws.amazon.com/message-queue/

Introduction to Message Queues



- Multiple producers and consumers can use the queue
- Messages are stored on the queue until they are processed and deleted
- Each message is processed only once by a single consumer (a.k.a. one-to-one or point-to-point communication)

source: https://aws.amazon.com/message-queue/

Message Queues – Advantages

- Promotes loosely coupled systems
- Provides fault tolerant solutions
- Scalable

Introduction to Publish/Subscribe Messaging

- Publish/Subscribe (Pub/Sub) messaging – asynchronous service-to-service communication used in distributed computing, (micro)services, and serverless architectures
- Use cases
  - enable event-driven architectures
  - decouple applications in order to increase performance, reliability and scalability

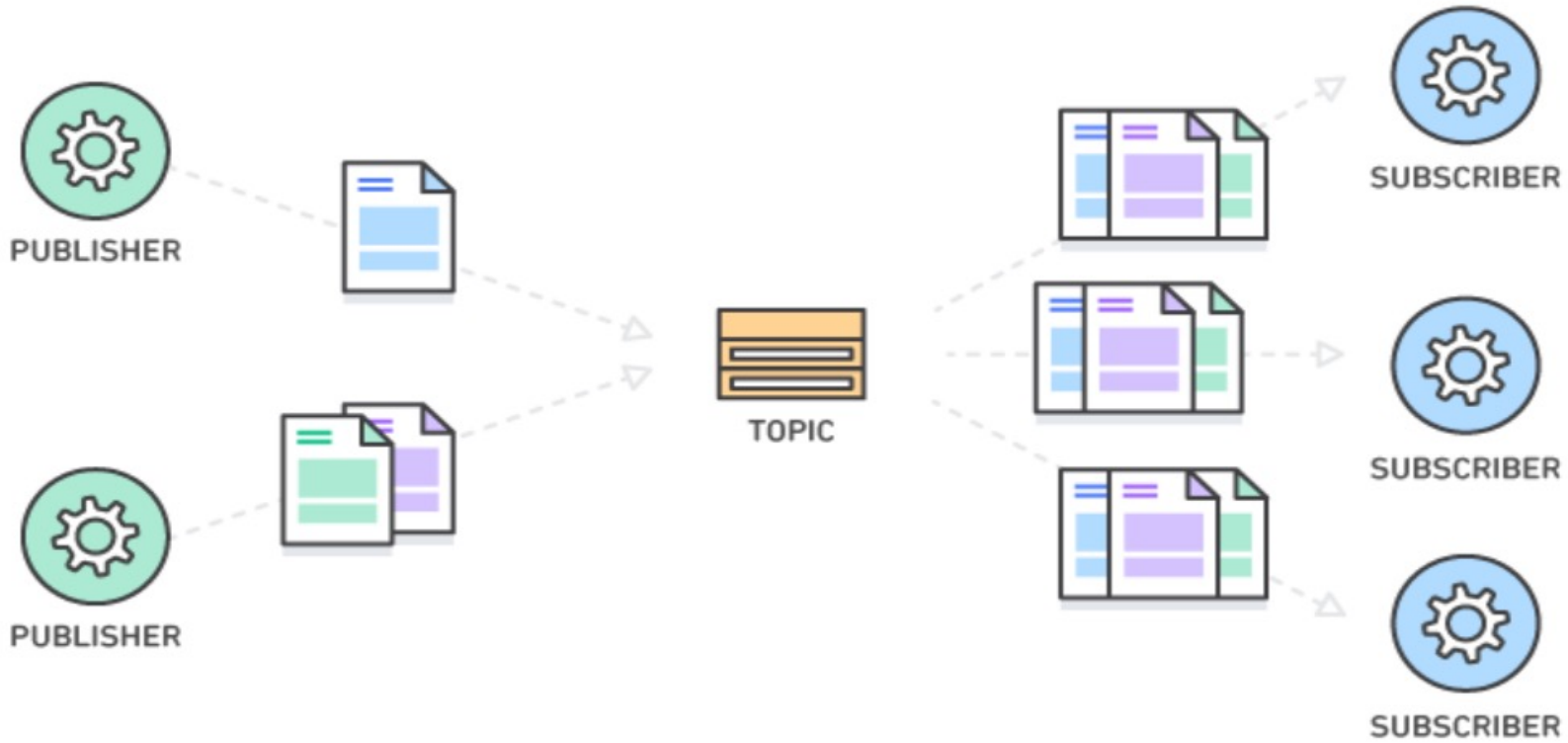
source: https://aws.amazon.com/pub-sub-messaging/

Introduction to Publish/Subscribe Messaging

- Publish/Subscribe messaging provides instant event notifications for distributed applications
- A Pub/Sub provides:
  - a lightweight mechanism to broadcast asynchronous event notifications
  - endpoints that allow software components to connect to the topic to send and to receive messages

source: https://aws.amazon.com/pub-sub-messaging/

Introduction to Publish/Subscribe Messaging



source: https://aws.amazon.com/pub-sub-messaging/

Introduction to Publish/Subscribe Messaging

- A message topics transfer messages with no (or very little) queuing, and push them out immediately to all subscribers
- All components that subscribe to the topic receive every message that is broadcasted, unless a message filtering policy is set by the subscriber
- The subscribers can process the message in parallel, each subscriber can perform different operations
- The publisher and the subscribers do not know each other

source: https://aws.amazon.com/pub-sub-messaging/

Message Queues Services – Examples

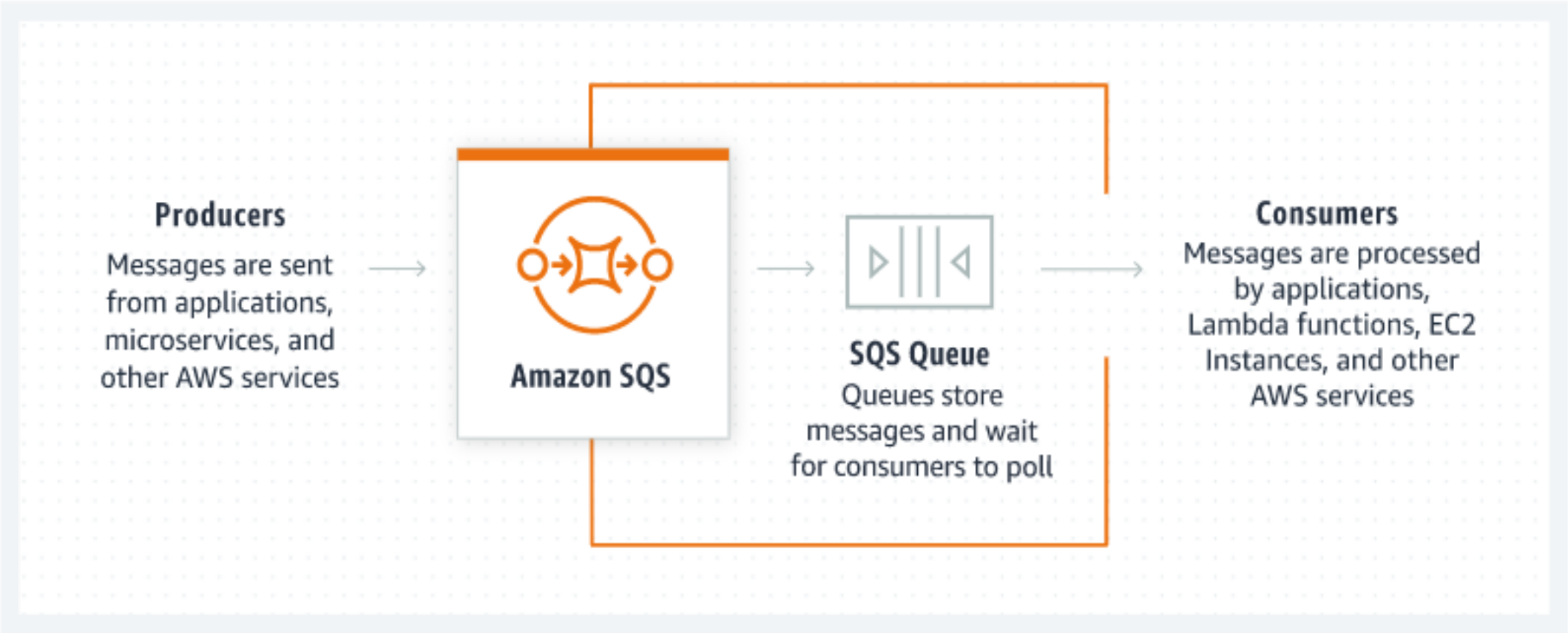
- Amazon Simple Queue Service (Amazon SQS)
- Microsoft Azure Storage queues and Service Bus queues
- IBM MQ
- Google Cloud Pub/Sub

Amazon Simple Queue Service (Amazon SQS)

- A fully managed message queuing service that enables to decouple and scale microservices, distributed systems, and serverless applications
- Deliver messages reliably
- Keep sensitive data secure
- Scale elastically
- Cost effective

source: https://aws.amazon.com/sqs/

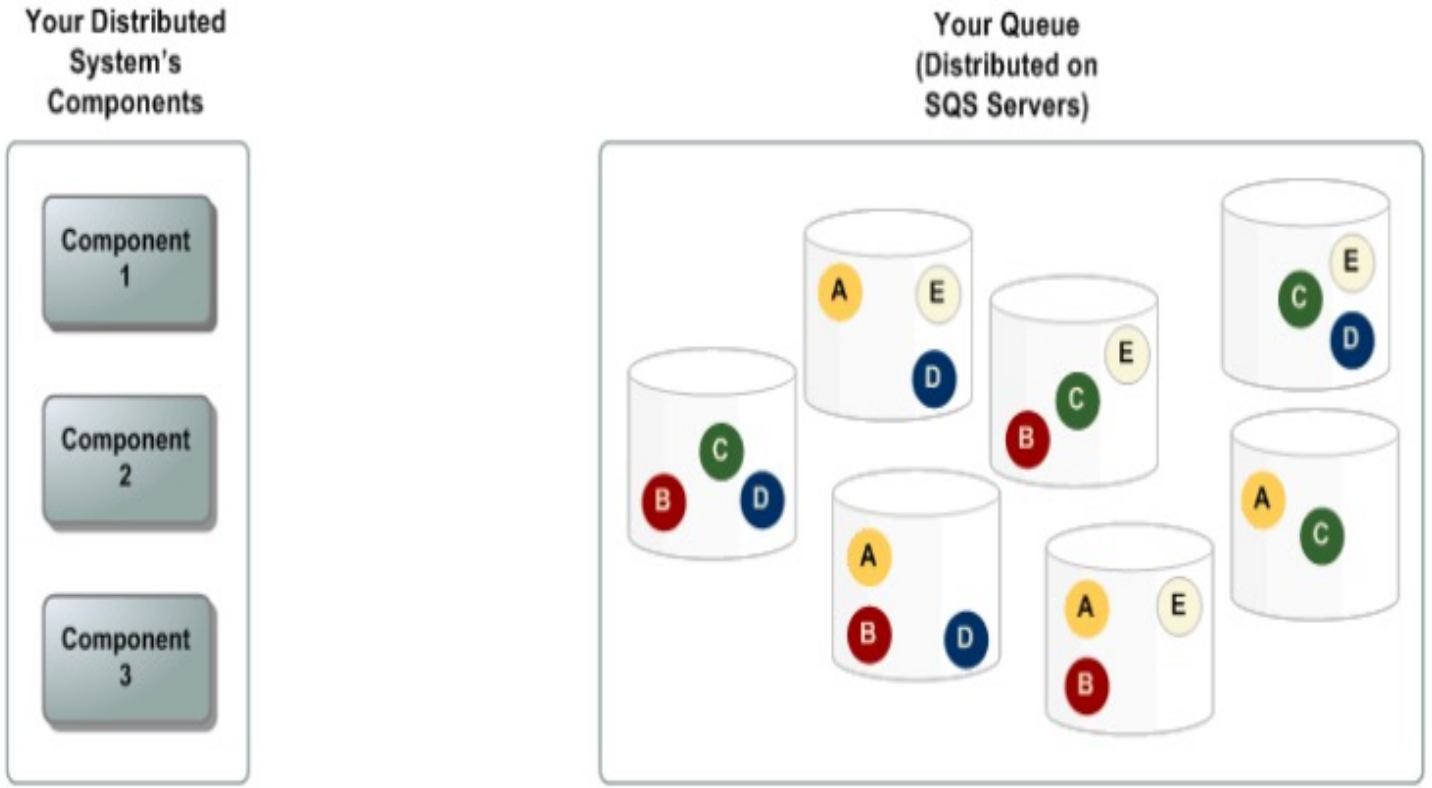
Amazon SQS



source: https://aws.amazon.com/sqs/

Amazon SQS – Basic Architecture – An Example

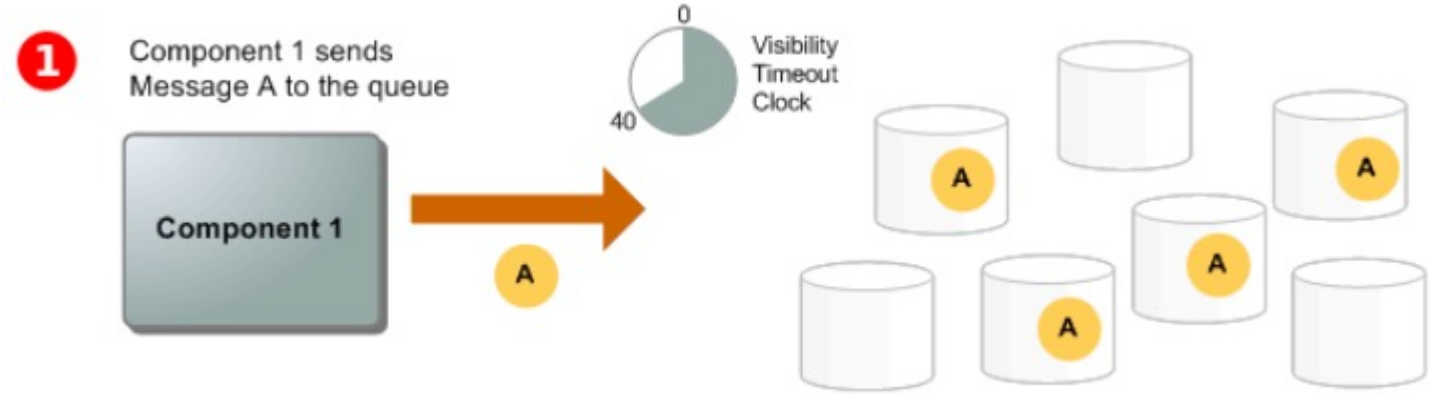
- Components – producers and consumers
- The queue stores the messages across multiple Amazon SQS servers



source (including images) : https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/sqs-basic-architecture.html/

Amazon SQS – Message Lifecycle – An Example

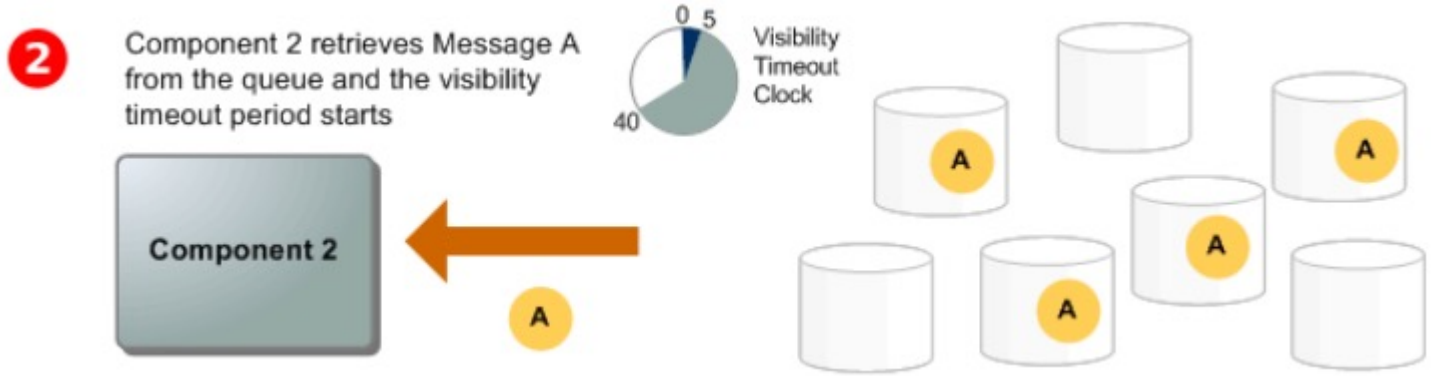
- The producer sends message A to a queue
- Message stored redundantly across Amazon SQS servers



source (including images) : https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/sqs-basic-architecture.html/

Amazon SQS – Message Lifecycle – An Example (contd.)

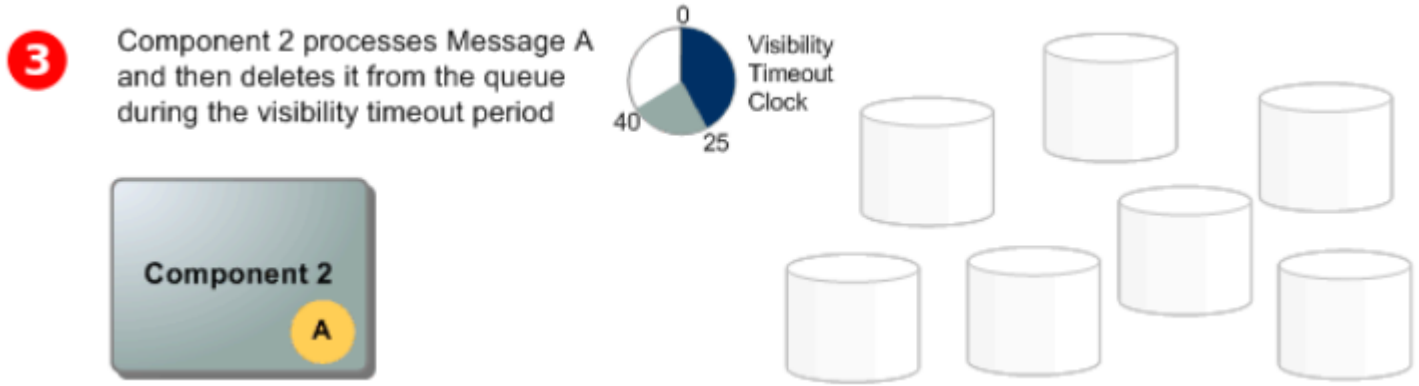
- While consumer processes a message, the message remains in the queue and it is not returned to subsequent receive requests for the duration of the visibility timeout



source (including images) : https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/sqs-basic-architecture.html/

Amazon SQS – Message Lifecycle – An Example (contd.)

- Once processed, the consumer deletes the message from the queue to prevent the message from being received and processed again



source (including images) : https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/sqs-basic-architecture.html/

Amazon SQS

Provides two types of queues

- Standard queue (default)
- First-In-First-Out (FIFO) queue

Amazon SQS – Standard queue

- Messages might be delivered out of order
  - e.g. Message 1, Message 2, Message 4, Message 3,…
- At-least-once message delivery (duplicated messages may be received)
  - e.g. Message 1, Message 2, Message 2, Message 3,…
- High throughput

source: https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/standard-queues.html

Amazon SQS – FIFO queue

- Messages are delivered in order
  - e.g. Message 1, Message 2, Message 3, Message 4,…
- Messages are only delivered once
  - e.g. Message 1, Message 2, Message 3, Message 4,…
- Limited throughput

source: https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/FIFO-queues.html

Amazon SQS – Use Cases

- Standard queue (default)
  - Decouple user requests from the processing
  - Allocate tasks to multiple worker nodes
  - Batch messages for future processing
- First-In-First-Out (FIFO) queue
  - Order of operations and events is critical
    - e.g. course enrollment, bank transactions

source https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/standard-queues.html

Amazon Simple Notification Service (Amazon SNS)

- A fully managed pub/sub messaging service for
  - application-to-application (A2A) communication
  - application-to-person (A2P) communication
- Provides many-to-many messaging between distributed systems, microservices, and event-driven serverless applications

source: https://aws.amazon.com/sns/

Amazon SNS – A2A



source: https://aws.amazon.com/sqs/

Amazon SNS – A2P



source: https://aws.amazon.com/sqs/

Amazon SNS

Provides two types of topics

- Standard topic
  - Use when message delivery order and possible message duplication are not critical
- FIFO topics
  - Use when must ensure strict message ordering and prevent message duplication

source: https://aws.amazon.com/sqs/

Resources

- Amazon SQS: https://aws.amazon.com/sqs/
- Amazon SQS API Reference:
- https://docs.aws.amazon.com/AWSSimpleQueueService/latest/APIReference/Welcome.html
- Amazon SNS: https://aws.amazon.com/sns/
- Amazon SNS API Reference: https://docs.aws.amazon.com/sns/latest/api/Welcome.html